

Exercise 4

What is intuitionistic propositional logic? How does the natural deduction for it differ from the system studied in the class? What are the applications of intuitionistic logic in computer science? Share your answer with your peers on the Moodle discussion forum.

Part 1: What is intuitionistic propositional logic?

Answer

Intuitionistic propositional logic (IPL) means: a formula is true only when we can build a proof of it.

Part 2: How does the natural deduction for it differ from the system studied in the class?

Answer

In classical logic, a proposition is either true or false, while is considered true only if there is a constructive proof of it.

Part 3: What are the applications of intuitionistic logic in computer science?

Answer

Applications include the Curry–Howard correspondence (proofs as programs), typed λ -calculus, proof assistants (e.g., Coq, Agda), program extraction, and verified software and protocols.